https://doi.org/10.1140/epjqt/s40507-026-00463-5

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EPJ Quantum Technology

RESEARCH Open Access

Quantum-driven enhanced machine learning algorithm for intrusion detection in Internet of things environment

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Abstract

Industry 4.0 and other advancements are made possible by an Internet of Things (IoT), one of the emerging technologies. In addition, security becomes a difficult problem when IoT devices communicate over wireless channels. It has recently been feasible to detect intrusion in an IoT environment using machine learning techniques. The amount of computation needed for training grows dramatically with huge volume of data. As the amount of data increases, the running time of many machine learning algorithms increases drastically. At the intersection of traditional machine learning and quantum computing, quantum machine learning promises to revolutionize the processing and analysis of large data sets by utilizing the special benefits of quantum physics. This work introduces Hybrid Quantum Neural Network (HQNN) contrived for overcoming these challenges and facilitate efficient Quantum enabled machine learning for real time intrusion detection systems. Real network traces are used to validate our proposed intrusion detection technology and our approach shows remarkable improvement in accuracy against conventional approaches.

Keywords: Industry 4.0; IoT; Machine Learning; Quantum computing; Hybrid Quantum Neural Network; Intrusion detection

1 Introduction

Industry 4.0 has garnered significant attention from interdisciplinary researchers, particularly in artificial intelligence and machine learning (AI/ML), and IoT, owing to its promising features. A key component of the Industry 4.0 Era's construction of smart cities, smart workplaces, and smart households is the IoT, which has emerged in recent years. These technologies hold the potential to enhance the economic efficiency of organizations. All IoT adoption, however, is thought to be particularly sensitive to security and privacy concerns. An effective way to reduce and respond to these cyber threats is to employ network intrusion detection system (IDS). This system detect and identify vulnerabilities in IoT based systems that pose a major threat to the security of any system connected to it as well as the overall smart environment [1]. Despite offering numerous benefits to end-users, security concerns persist. The widespread adoption of AI/ML, coupled with its unique

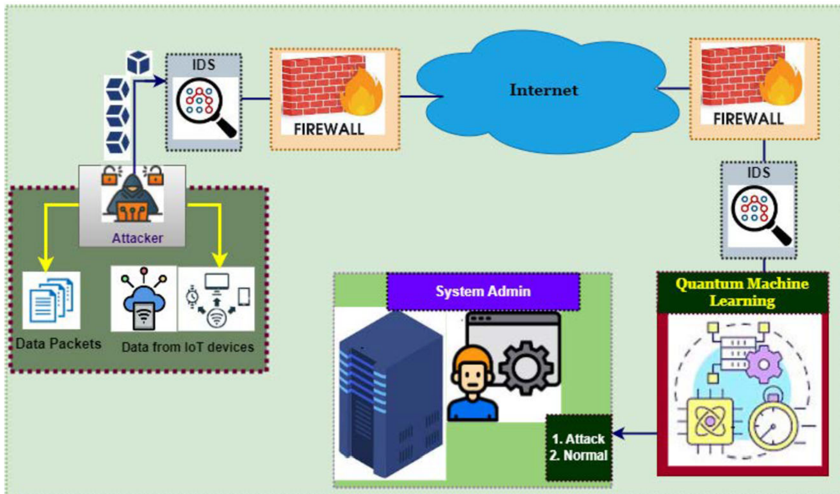


Figure 1 End-to-end network security framework illustrating how intrusion attempts are detected through IDS and firewall layers, where the filtered data is then evaluated by a Quantum Machine Learning model to support attack classification

capabilities, presents an opportunity to address security challenges in real-time applications such as healthcare, finance, and cloud computing. As shown in Fig. 1, network packets, sensor readings, device activity logs, and other pertinent data can be gathered from IoT devices. The IDS can be dispersed throughout the network or centrally installed on network gateways, edge devices, firewalls, and other infrastructure, depending on the deployment scenario. Continuous measurement of network traffic behavior is essential to detect unusual activities, prompting network administrators to implement counter measures when such behavior is identified [2]. In IoT networks, data theft during transmission is a significant concern. Classical artificial intelligence techniques play a crucial role in detecting such intrusions. However, the training time of classical learning approaches scale with the dataset size [3], leading to delay in deploying IDS systems in real IoT environments. Consequently, IDS systems may not respond promptly to threats, particularly in time-critical applications. The known limited processing power and storage volume of IoT devices, classical learning-based approaches for IDS are not optimal.

Quantum machine learning algorithms address the limitations inherent in traditional learning algorithms utilized in IDS. Quantum computers utilize quantum phenomena like superposition to achieve computation speed faster than classical computers, offering the possibility of accelerating the training process for machine learning models [4]. With quantum computing, complex optimization tasks involved in training machine learning models can be executed efficiently, to reduce the time required for training. These algorithms may enable more accurate and faster detection of anomalies and intrusions within IoT network traffic [5]. Additionally, the inherent parallelism of quantum computing could allow for the simultaneous processing of multiple data streams, improving the scalability of IDS to handle large volume of network traffic in real-time. Though quantum computing in its evolving stage of development but still practical quantum computes are act as an expert when compared with classical computers in the area of machine learning. Research and development efforts are ongoing to overcome technical challenges and scale up quantum computing capabilities. In summary, quantum computing shows promise for addressing the time-related limitations of machine learning in IDS by potentially speeding up the training process, improving classification accuracy, and enabling real-time processing of large volumes of network data.

Contribution of the paper:

- This work proposes the integration of classical feature inputs with their quantum counterparts to obtain a hybrid classical quantum framework suitable for intrusion detection systems.
- This approach integrates classical input features with quantum computational circuits to enhance performance of the intrusion detection system.
- The PCA based dimensionality reduction algorithm is used to extract essential features from heterogeneous IDS dataset and encoded using an efficient Quantum feature mapping technique.
- The approach utilized repetition code component for the detection and correction of errors that may occur during Quantum processing
- The proposed methods are evaluated on real-time datasets, considering key performance metrics such as Accuracy, Precision, Recall, and F1score.
- Experimental results obtained on IBM's Quantum simulator and real Quantum computing platforms shown significant enhancements in IDS detection performance.

The rest of the paper is organized as follows: Sect. 2 presents a comprehensive review of the existing literature relevant to intrusion detection and quantum–classical hybrid learning approaches. Section 3 explains the architecture of the proposed system in detail, including the feature selection method, model framework, and workflow. Section 4 provides a complete description of the datasets used in this study, highlighting their characteristics, preprocessing steps, and feature extraction process. Section 5 reports the discussion of the experimental results and performance comparison. Finally, Sect. 6 summarizes the key findings of the study and outlines potential directions for future research.

2 Literature review

The IDS stands as a pivotal component in identifying internet threats, capable of detecting attacks originating from either network or host sources. The resource constraint protocol (RPL) for 6LoWPAN network is vulnerable to destination information solicitation (DIS) flooding attacks, which degrade network efficiency by increasing control packet overhead and resource demand. To address this, a defense system called DIS mitigation in RPL is suggested [6], effectively identifying and mitigating such attacks without significant overhead. Due to IoT technology, security of IoT devices becomes challenging. The traditional approaches like Rivest-Shamir-Adleman (RSA) and Advanced Encryption Standard (AES) will not be suitable for quantum computing environment [7]. The author highlights [8] the potential use of quantum computing in enhancing IoT security for IoT devices. This work proposed AdaBoost with Boruta feature selection, which performs better in terms of accuracy compared to other IDS system. To classify botnet based attacks, an efficient and robust model was developed by [9]. The author [10] proposed a hybrid learning approach for spotting attacks in IoT devices. The model was evaluated using publicly available datasets, attaining a training and testing accuracy of 93.5% and 92.53% respectively. The author [11] proposed ML-based IDS to familiarize the advancements in IoT technology. Additionally, [12] introduced a cost-effective semi-supervised IDS system aimed at mitigating issues such as low accuracy and high detection time. The system was developed using a modified random forest algorithm and was trained on the UNSW-NB15 dataset. An effective and robust ML-based IDS to safeguard the Internet of Medical Things (IoMT) network is developed by [5] and this model utilized a combination of Grey Wolf Optimization (GWO) and Particle Swarm Optimization (PSO) for feature selection. The experiment was conducted using KDD and BoT IoT datasets, achieving an accuracy of 94.66% with classifiers including K-nearest neighbor, Naïve Bayes, support vector machine, artificial neural network, and decision tree. An improved IDS that utilizes gradient boosting (GB) and decision tree (DT) algorithms via Catboost for IoT security was introduced in [13]. This model was evaluated using multiple publicly available datasets, such as IoT, demonstrating an accuracy of 92.3%. A lattice-based hash troop is employed for wireless IDS in conjunction with a machine learning approach [14]. The deep neural network analyzes various features from IoT datasets and demonstrates high accuracy.

Intrusion detection using gravitational search algorithm as a feature selection and quantum Neural Network model for classification was proposed in [15]. This model enhanced the effectiveness of intrusion detection by employing Z-score normalization technique. Quantum computing enhances encryption, federated learning preserves privacy, and 6G facilitates real-time secure data processing. The paper [16] proposed a framework for quantum computing by integrating with federated learning especially for 6G networks. This approach is further enhanced to support the features of 7G network and to shape the future of IoT security. Implementing quantum learning strategies becoming important, particularly in cyber security, where machine learning enhances malware detection effectiveness, scalability, and action-ability. Managing cyber security risks associated with quantum learning was essential to overcome the obstacles effectively. Deep learning, support vector quantum, and Bayesian classification are promising quantum learning and statistical technologies for mitigating cyber attack consequences. Intelligent security systems must uncover hidden patterns within network data and develop data-driven quantum learning models to counter threats effectively. A novel variational quantum multi-class classifier, using qubits to represent N labels and optimizing circuit parameters based on fidelity between true and output label states is proposed. By reducing circuit width and auxiliary particles, it achieved high accuracy of 91.8% for 4 classifications and 92% for 8 classifications on MNIST, 85.3% for 8 classifications on CIFAR-10, and 76% for 16 classifications on CIFAR-100 datasets through simulation experiments [17]. For both unsupervised and supervised learning, quantum search algorithm was proposed in [18]. These algorithms were used to determine the class membership of quantum systems without prior knowledge of their states. Additionally, a general scheme was developed for evaluating dependability between states of unknown quantum systems. The supervised time-series classification, employing hybrid quantum-classical machine learning techniques was proposed in [19]. This method utilized Hamiltonian kernel to capture pairwise temporal relationships between instances. By optimizing the kernel weights through multiple kernel learning, the approach called hybrid quantum-classical-convex neural network, termed QCC-net was used. A quantum convolutional network used three-qubits quantum circuit for image was proposed in [20]. By assessing expressibility and entangling capability, this approach demonstrated superior performance on MNIST, Fashion MNIST, and Iris datasets.

A novel approach namely, continuous variable quantum RNN was proposed in [21] to improve time series prediction efficiency. By integrating continuous-variable quantum resources into recurrent neural networks, their approach reduces complexity and energy consumption. The quantum convolution neural networks as a quantum analogue of CNN for classical data classification was proposed in [22]. By utilizing two qubits interactions and shallow depth quantum circuits, QCNN achieved remarkable classification accuracy on MNIST and Fashion MNIST. Various QCNN models were evaluated based on circuit structures, data encoding methods, pre-processing techniques, cost functions, and optimizer, outperforming CNN models under similar training conditions. Notably, the QCNN algorithm's compatibility with Noisy Intermediate-Scale Quantum (NISQ) devices makes it promising for practical implementations in quantum machine learning. Most of the intrusion detection system uses the quantum neural networks for optimizing their performance within current quantum computing constraints. The approach included classical feature encoding, QNN classifier selection, and performance tuning to maximize current quantum computational power. Quantum Cat Swarm Optimization based Clustering (QCSO) with intrusion detection technique was proposed for IoT environments, aiming at energy efficiency and security [23]. This model utilized QCSO algorithm to select cluster heads and organized clusters, considering energy efficiency, cluster distance of inter and intra, and node density. Intrusion detection was achieved through a Harmony Search Algorithm with Recurrent Neural Network approach, optimizing the hyperparameters. Experimental analysis validates the efficiency of the proposed work compared to existing models, showcasing promising results in IoT environments.

The author [24] proposed hybrid artificial neural network combined CNN and LSTM RNN models for feature extraction and sequential data analysis, by achieving 80% of accuracy across various attack types. Additionally, quantum algorithms QSVM and VQC were explored, achieving 60% accuracy with minimal features, showing potential for standalone intrusion detection in large data centers. Through data transformation techniques, network intrusion system was effectively classified and predicts the zero-day attacks.

3 Materials and methods

Today, Quantum Machine Learning (QML) has emerged as a prominent trend in quantum computing, using quantum processing to execute computationally complex routines on quantum devices. In classical computing, information is processed using binary digits, or bits, which can take values of either 0 or 1. However, in quantum computing, the basic unit of information is the quantum bit, or qubit. The Qubits $|0\rangle$ and $|1\rangle$ are represented as state vectors within a mathematical space known as Hilbert space ($\mathcal{H}$) as given in the Equation (1). The states of qubits in a quantum computer are described by complex probability amplitudes. When a quantum state is observed, the probabilities of different outcomes are determined by the squared moduli of these amplitudes. This means that the probability of observing a particular state is the square of the magnitude of its probability amplitude as given in the Equation (2):

$$|\phi\rangle = a_0|0\rangle + a_1|1\rangle; |0\rangle, |1\rangle \in \mathcal{H} \tag{1}$$

$$a_0, a_1 \in \mathbb{C}; |a_0|^2 + |a_1|^2 = 1 \tag{2}$$

The data obtained from the three IoT security datasets consists of high dimensional feature vectors generated from diverse sources of network traffic lead to significant computational overhead while model training. The large feature space directly influences the training time and enhances the complexity of classical machine learning algorithms. For

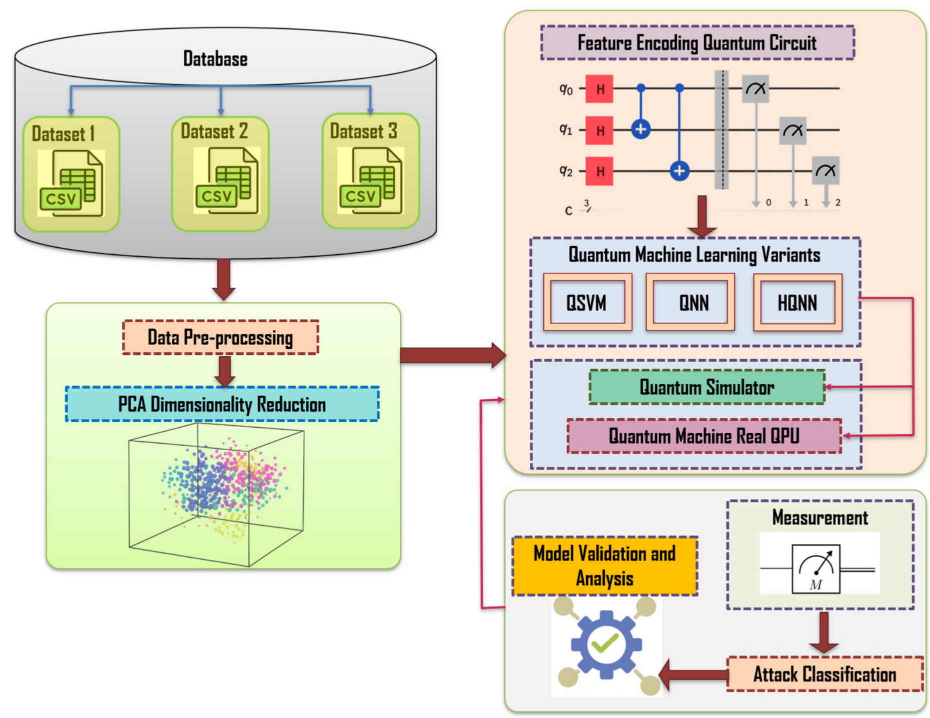


Figure 2 Overall workflow of the proposed Quantum Machine Learning–based intrusion detection system

this purpose, Principal Component Analysis (PCA [25]) is applied in order to extract the most informative components, while reducing redundant and correlated features, thereby minimizing model training overhead. However, the reduced feature vectors cannot be directly interpreted by quantum processors. To bridge this gap, the selected PCA features are transformed using the ZZFeatureMap, Quantum feature-encoding technique embedding classical numerical values into entangled quantum states. These encoded quantum features are then fed into quantum machine learning classifiers, including the Quantum Support Vector Machine (QSVM), Quantum Neural Network, and the proposed HQNN. These models are executed both on a state vector simulator and on real quantum hardware, and the final classification decision is derived from the measurement outcome of the quantum circuits. The overall architecture of the proposed quantum classical intrusion detection framework is illustrated in Fig. 2.

4 Dataset description

In this study, three heterogeneous and widely benchmarked miscellaneous well-known IoT and network security datasets Canberra BoT-IoT Dataset [26], the ACI IoT Network Traffic Dataset [27] and the ROUT-4-2023 RPL-Based Routing Attack Dataset [28] are utilized to assess the performance of the proposed Quantum classifier framework. These datasets collectively cover flow level, packet-level, and routing-level attack behaviors, hence ensuring a robust Evaluation of the model's adaptability across complex and diverse intrusion environments. Every dataset is characterized by high-dimensional feature spaces and large traffic volumes. making them suitable for quantum machine learning research aimed at detecting IoT cyberattacks with improved precision, scalability, and generalization.

Table 1 Summary of benchmark IoT datasets used in this study

Dataset Name	No. of Features	Total Samples
BoT-IoT	26	1,048,575
ACI IoT	76	1,048,575
ROUT-4-2023	39	494,021

4.1 BoT-IoT dataset description

The first benchmark dataset used in this work is the BoT-IoT Dataset generated by Cyber Range Lab at the UNSW Canberra Cyber Centre [26]. The dataset was provided from a realistic IoT enabled Network environment by integrating benign IoT device behavior with large scale Botnet driven malicious activities. This dataset was made available in different formats such as raw pcap captures, Argus flow files, and CSV formatted records, allowing for flexible pre-processing and Analysis. BoT-IoT dataset comprised of 1,048,575 traffic instances as depicted in Table 1. Each instance was described by 26 features. The information are to be captured includes packet statistics, flow duration, protocol usage, and TCP flags. counts, service ports, and behavioral traffic indicators. The dataset includes a wide range of attack types including Distributed Denial of Service (DDoS), Denial of Service (DoS) and reconnaissance scans, information theft, and key logging events. Its realistic traffic distribution and broad Coverage of high-intensity attack patterns makes it a highly valued benchmark for evaluating quantum-enhanced intrusion detection models.

4.2 ACI IoT network traffic dataset description

The second dataset used for this study is the ACI IoT Network Traffic Dataset. In particular, the data set cited as [27] was specifically constructed by Army Cyber Institute (ACI) to support development in advanced machine learning and IoT network intrusion detection systems driven by AI. It fills a critical gap by providing a large scale, diverse, and realistic IoT traffic repository. The dataset includes 1,048,575 samples, each described by 76 high dimensional features, represent flow based statistics, packet-level metadata, and time based traffic characteristics, application-layer attributes, and IoT-specific communication patterns, including MQTT, CoAP, and HTTP(S) traffic. The dataset captures both IoT behavior and multiple categories of malicious traffic such as botnet activity (Mirai and Gafgyt), flooding attacks, slow-rate communication disruptions, protocol exploitation, and anomalous device-to-cloud interactions. This dataset is highly suitable for training quantum–classical learning models due to its heterogeneity, feature richness, and balanced representation of real IoT operational states.

4.3 ROUT-4-2023 RPL-based routing attack dataset description

The third dataset incorporated in this research is the ROUT-4-2023 RPL-Based Routing Attack Dataset [28]. It focuses explicitly on routing layer intrusion behavior in IoT networks operating under the RPL (Routing Protocol for Low-Power and Lossy Networks) protocol, the dominant routing protocol in modern IoT deployments. The dataset was generated using the Cooja network simulator in the Contiki-NG environment, ensuring accurate emulation of constrained IoT devices and routing behavior. ROUT-4 contains 494,021 labeled traffic samples represented by 39 features that describe RPL control messages, rank values, parent selection patterns, network topology updates, and routing inconsistencies. The dataset includes four major routing attack classes namely Blackhole

Attack, Flooding Attack, DODAG Version Number Attack and Decreased Rank Attack. This dataset is particularly useful for evaluating the detection performance of quantum machine learning models in routing-layer intrusion scenarios that are difficult to detect using conventional flow-based techniques. Table 1 provides a consolidated overview of the three datasets utilized in this study, highlighting the number of features and total traffic instances available for training and evaluating the proposed model.

4.4 Dimensional reduction using PCA

The scarcity of qubits presents a major obstacle in the field of near-term quantum devices. Hence there exists an essential need to decrease the feature dimensions of the benchmark datasets to make them suitable for use while feeding to actual quantum systems. It's crucial to remember that a classifier's performance on a smaller dataset is mostly dependent on the features used. In real-time scenario, the classifier performance depends on the selection of significant features. Therefore, it is crucial to ensure that the chosen features are relevant for enhancing the classifier's performance. Among various feature selection approaches, PCA stands out as a promising method for identifying the most informative features from intrusion detection datasets. PCA is applied on all the three datasets to reduce the feature dimensions of the original dataset, transforming the incoming data into a more manageable format. It selects only the most valuable features from the extensive dataset, thereby ensuring the accuracy of the classifier model. By utilizing PCA, over-fitting is minimized, and classification overhead is reduced.

The pre-processing and labeling of the datasets are performed using the Scikit-learn machine learning package from the Python library. Subsequently, the dataset is normalized using the standard scalar function. To boost classifier accuracy and reduce training time, a large dataset is typically given to the PCA algorithm for dimension reduction. In the proposed system, 70% of the dataset is allocated for training the model, while the remaining 30% is reserved for testing and validating the model. This preparation step ensures that the data are in an appropriate format for PCA analysis and helps to improve the effectiveness of the feature selection process. As a part of pre-processing stage each feature x_i in the dataset is standardized using the Equation (3):

$$x^{std} = \frac{x_i - \sigma_i}{\delta_i} \tag{3}$$

where σ_i and δ_i represent the mean and standard deviation of feature vector. Then the co-variance is computed using the Equation (4):

$$\sum = \frac{1}{n}\sum_{i=1}^{n} x_i^{std}.(x_i^{std})^T \tag{4}$$

Then eigen value (λ) and vectors (ν) are computed from the co-variance matrix given in the Equation (5):

$$\sum \nu = \lambda \nu \tag{5}$$

The eigen values are arranged in decreasing order according to the eigen vectors. Select the top k eigenvectors conforming to the highest k eigenvalues to form the matrix called

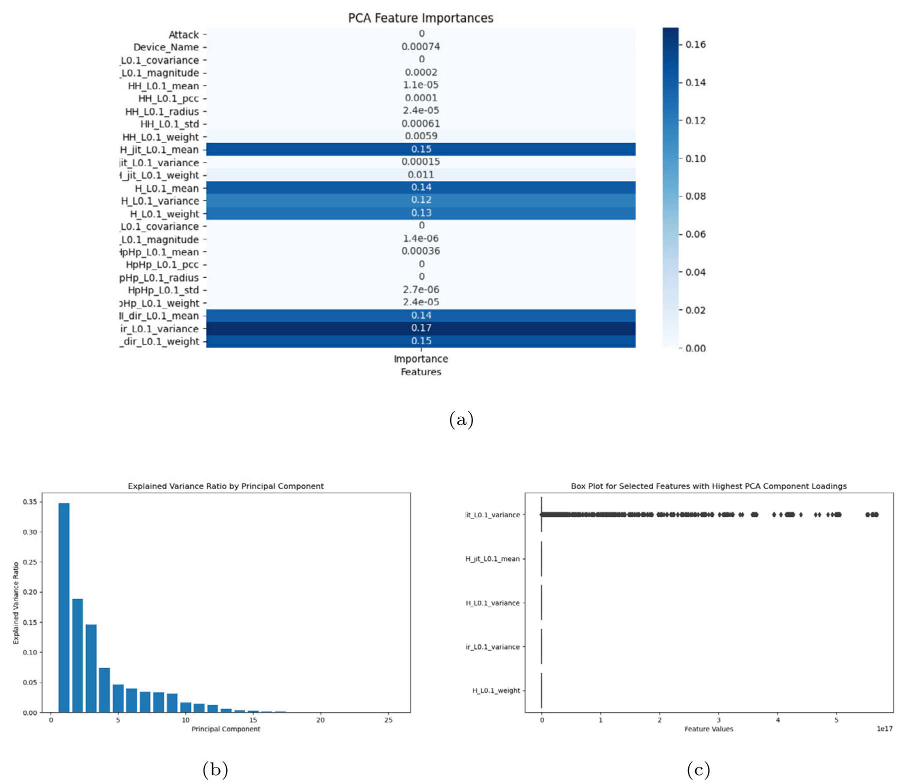


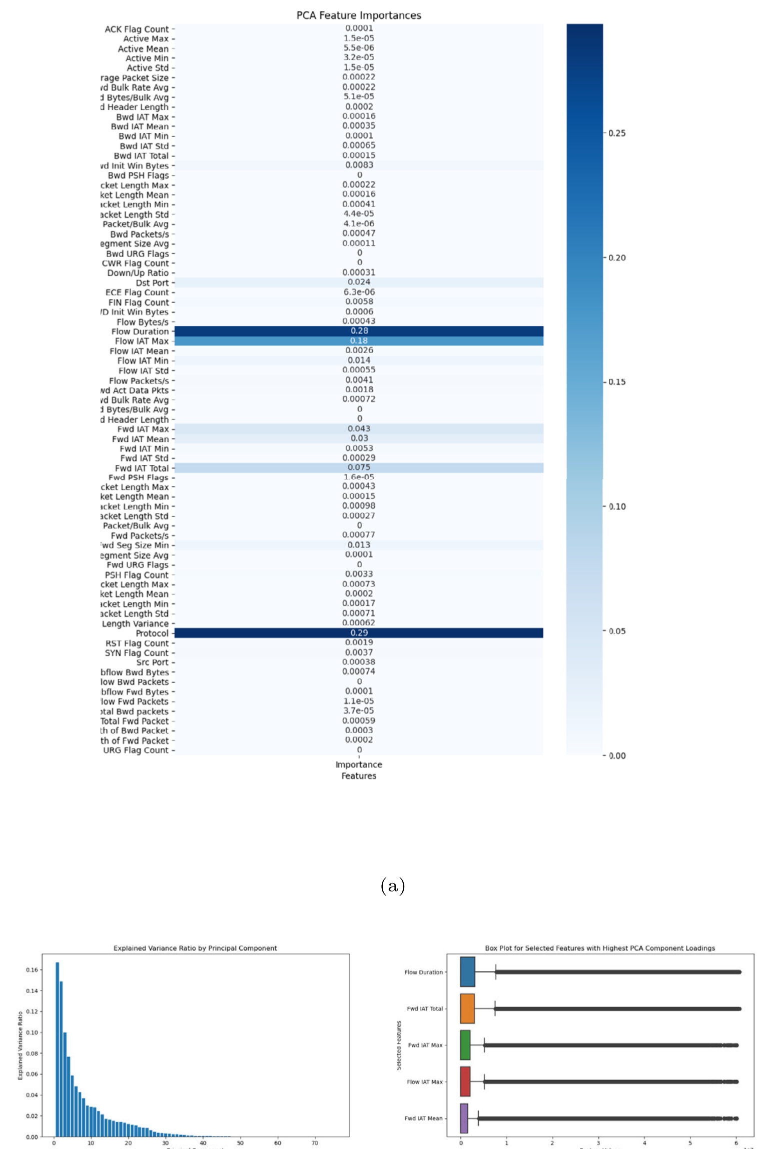
Figure 3 Visualization of the principal components extracted from IoT-BoT dataset. (a) Heatmap showing the strongest PCA derived feature correlations. (b) Histogram of various component distribution from PCA. (c) Box plot for the significant features extracted

principal component V_k. Project the standardized data X^{std} onto the selected principal components as given in the Equation (6):

$$X_{reduced} = X^{std}.V_k \tag{6}$$

where, $X_{reduced}$ represents reduced features.

Figure 3 show the distribution of principal components generated by applying PCA on the IoT-BoT dataset. After applying PCA, the contribution of each original feature to the principal components was analyzed to identify the most dominant features. Based on the PCA results obtained for different values of k, the first principal component was chosen as it captures the highest proportion of variance. The features with the highest loading values on this component were identified as the most influential as shown in Fig. 3(a) and therefore considered as potential features for classification. Multiple trials were conducted by varying k from 1 to 8, and it was observed that $k = 5$ provides the optimal balance between variance retention and model performance. This selection ensures that the reduced feature set preserves the essential information required for accurate intrusion detection while effectively minimizing dimensionality. Figure 3(b) illustrates the variance captured by each principal component. As shown in Fig. 3(c), the first principal component that retains the highest variance and highlights the most influential features (HH_jit_L0.1_mean, H_L0.1_mean, H_L0.1_weight, Ml_dir_L0.1_variance, and Ml_dir_L0.1_weight) among the original features.



(a)

(b) (c)

Figure 4 Visualization of the principal components extracted from the ACI dataset. (a) Heatmap showing the strongest PCA-derived feature correlations. (b) Histogram of various component distribution from PCA. (c) Box plot for the significant feature extracted

Figure 4 show the distribution of principal components generated by applying PCA over the ACI dataset. After applying PCA, the contribution of each original feature to the principal components was analyzed to identify the most dominant variables. Based on the PCA results obtained for different values of k, the first principal component was chosen as it captures the highest proportion of variance. The features with the highest loading values on this component were identified as the most influential as shown in Fig. 4(a) and therefore considered as potential features for classification. Multiple trials were conducted by varying k from 1 to 8, and it was observed that $k = 5$ provides the optimal balance between variance retention and model performance. This selection ensures that the reduced

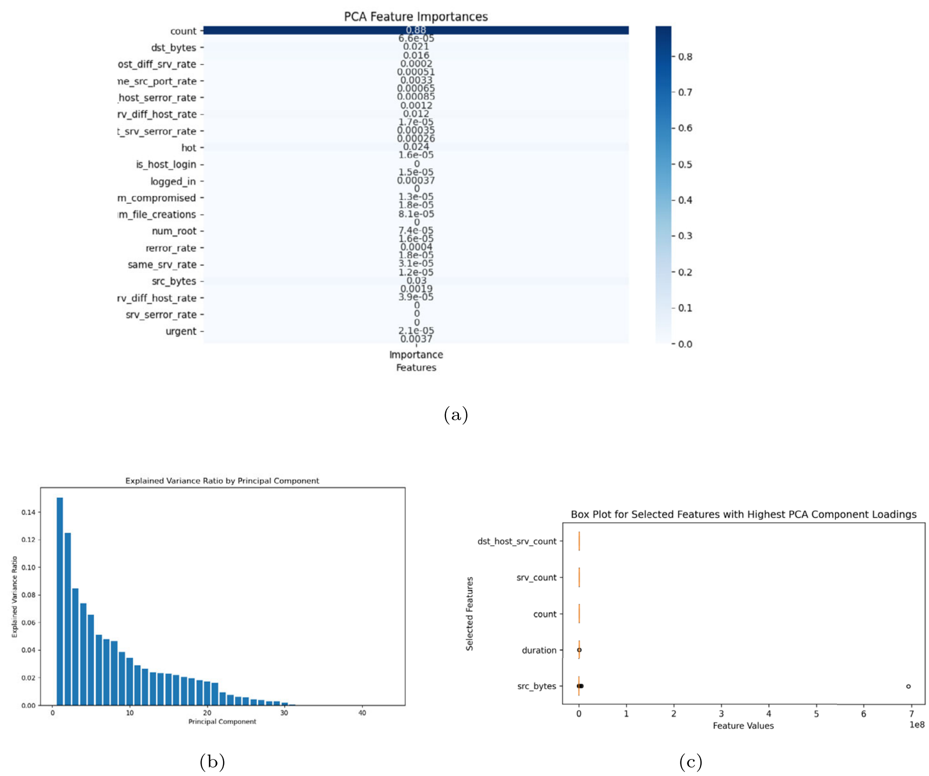


Figure 5 Visualization of the principal components extracted from the KDD dataset. (a) Heatmap showing the strongest PCA-derived feature correlations. (b) Histogram of various component distribution from PCA. (c) Box plot for the feature extracted

feature set preserves the essential information required for accurate intrusion detection while effectively minimizing dimensionality. Figure 4(b) illustrates the variance captured by each principal component. As shown in Fig. 4(c), the first principal component retains the highest variance and highlights the most influential features (Flow Duration, Flow IAT Total, Fwd IAT Max, Flow IAT Max and Fwd IAT Mean) among the original features.

Figure 5 show the distribution of principal components generated by applying PCA over the KDD dataset. After applying PCA, the contribution of each original feature to the principal components was analyzed to identify the most dominant variables. Based on the PCA results obtained for different values of k, the first principal component was chosen as it captures the highest proportion of variance. The features with the highest loading values on this component were identified as the most influential as shown in Fig. 5(a) and therefore considered as potential features for classification. Multiple trials were conducted by varying k from 1 to 8, and it was observed that $k = 5$ provides the optimal balance between variance retention and model performance. This selection ensures that the reduced feature set preserves the essential information required for accurate intrusion detection while effectively minimizing dimensionality. Figure 5(b) illustrates the variance captured by each principal component. As shown in Fig. 5(c), the first principal component retains the highest variance and highlights the most influential features (dst_host_srv_count, srv_count, count, duration and src_bytes) among the original features. This concludes that PCA effectively concentrates the meaningful variance into a lower-dimensional representation,

Algorithm 1 Feature Encoding Algorithm for Quantum Space

Require: Classical dataset $D = \{x_i, y_i\}$ where x_i and y_i represent feature of ith data point and $0 < i < n$
Require: Number of qubits n in the quantum circuit
  Initialize: Create a quantum circuit with n qubits
  for Each x_i in dataset **do**
    for Each $x_{i,j}$ feature of data point **do**
      Apply Hadamard gate to qubit j
      For each pair of features $x_{i,j}$ and $x_{i,k}$
      Apply controlled-rotation between qubits j and k
      Determine rotation angle $\theta_{j,k}$ based on feature values
      Apply controlled-rotation using $\theta_{j,k}$ as rotation angle
    end for
  end for
  Measure qubits in the circuit
  Perform classical post-processing on measurement outcomes

enabling faster training and reducing overfitting without losing predictive performance. The features identified by PCA, exhibit the highest variance are selected for further processing on the quantum computer.

4.5 Encoding classical data into quantum space

Data encoding techniques in quantum machine play an important role in representing classical data into quantum states. Building a classification model is made possible by mapping a data point to a quantum state provided by efficient ZZFeatureMap. In particular, it makes it easier to construct a kernel function that uses the Hilbert–Schmidt inner product to evaluate the similarity between two data points, x_i and x_j projected in the Hilbert space. The algorithm 1 describes the steps involved in ZZFeatureMap algorithm, from encoding classical data into quantum states to perform post processing for machine learning tasks. To identify the optimal k, an empirical evaluation was conducted by varying the number of retained principal components from $k = 1$ to 8 and recording the corresponding classification accuracy. The Fig. 6 illustrates the effect of PCA dimensionality K varied from 1 to 8 dimensions on the classification accuracy of benchmark datasets, providing a comparative view of accuracy variations and highlighting the saturation point for each dataset. The optimal value of k, marked in yellow in the Fig. 6, shows that accuracy begins to stabilize above $k = 5$ across all datasets. Although a slight improvement in accuracy is observed for $k \geq 6$, the gain is minimal only marginal decimal level increments. Therefore, $k = 5$ provides the best trade-off between accuracy and computational cost, and is selected as the optimal qubit requirement for all three benchmark datasets, as it consistently delivers high accuracy without incurring unnecessary qubit overhead.

Figure 7 illustrates the 5-qubit quantum feature map used for encoding the significant PCA selected features of all the three benchmark datasets. Each qubit is initialized with a Hadamard gate followed by parameterized phase rotation gates representing the nonlinear mapping of those significant features. Controlled-phase interactions between qubits capture higher-order correlations among the selected features. This feature map is used as

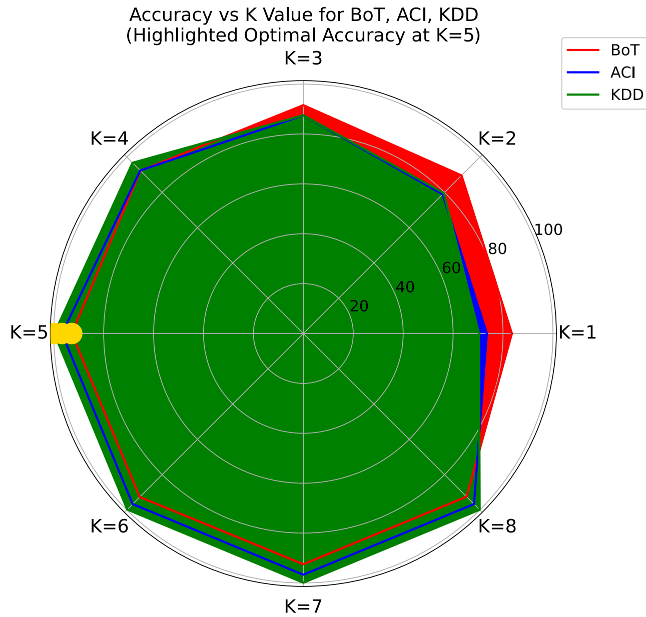


Figure 6 The impact of PCA dimensionality (K = 1–8) on classification accuracy for benchmark datasets

the input embedding circuit for the downstream QML classifier. These encoded features find application in both quantum simulation and quantum computation.

4.6 Error analysis from quantum processing using repetition codes

Errors that may arise during quantum processing can be found and fixed via repetition codes. The proposed approach adopted repetition codes due to its simplicity, robustness against the qubits error and scalability. By redundantly encoding information over several qubits. This enhances the precision and dependability of quantum computations and algorithms. The repetition codes is applied after the input features are encoded to ensure error free qubits in quantum processing. In our experimentation, the repetition codes are implemented for each dataset (BoT, ACI and KDD) after encoding features using ZZFeature mapping. The Fig. 8(a), 8(b), 8(c) shows the result of repetition codes applied for BoT dataset with different trials of 80, 100 and 120 respectively. The results indicated that 97% of success rate in terms of its performance in error correction due to its noise in quantum processing. The Fig. 9(a), 9(b), 9(c) shows the result of repetition codes applied for ACI dataset with different trials of 80, 100 and 120 respectively. The results indicated that 97% of success rate in terms of its performance in error correction due to its noise in quantum processing. The Fig. 10(a), 10(b), 10(c) show the results of repetition codes applied for KDD dataset with different trials of 80, 100 and 120 respectively. The results indicated that 97% of success rate in terms of its performance in error correction due to its noise in quantum processing.

4.7 Quantum Support Vector Machine (QSVM)

Support Vector Machine (SVM) is a legendary form in Machine Learning used to categorize dossier into labels. SVM will be the optimal hyperplane for the given datasets. In order to perform the optimization, ML model must minimize the value of w with respect

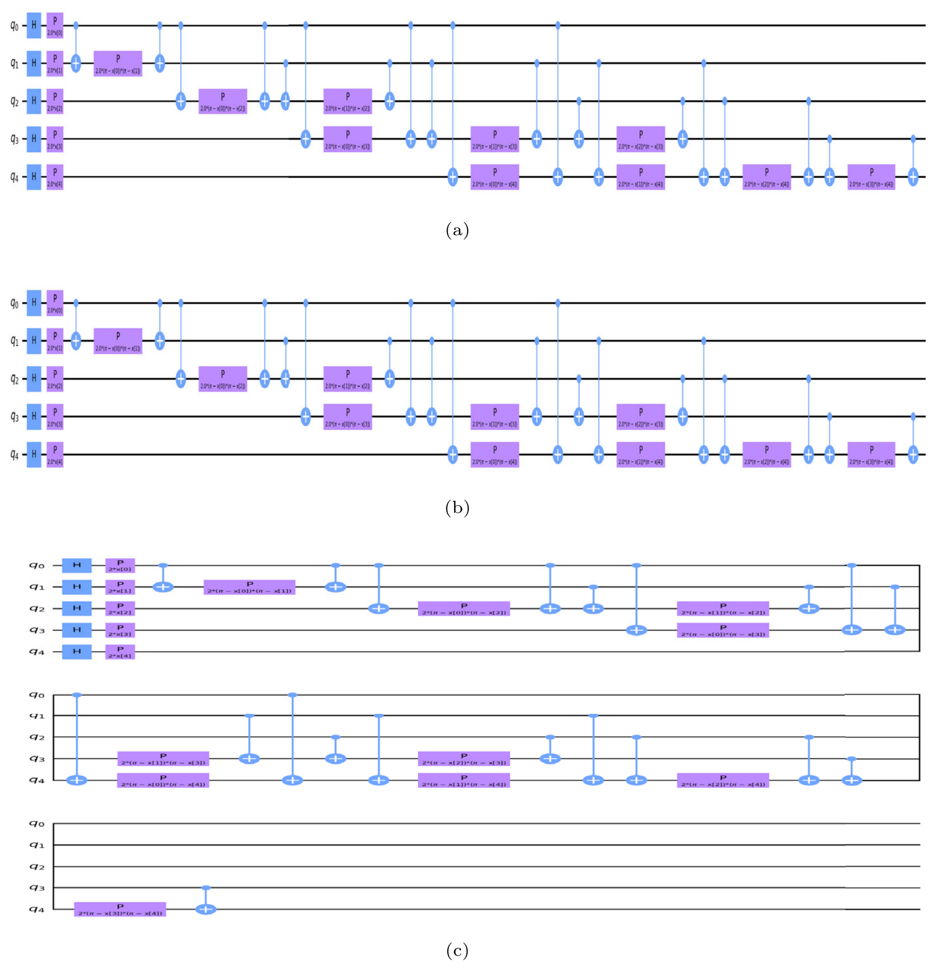


Figure 7 5-qubit Quantum ZZFeaturemaps encoding the PCA selected significant features for all the three benchmark datasets: (a) BoT, (b) ACI, and (c) KDD

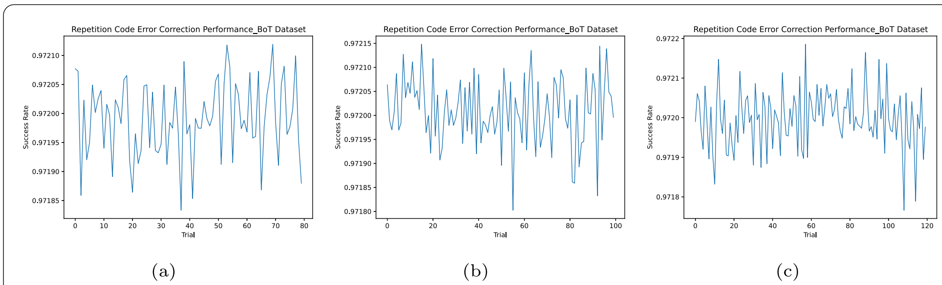
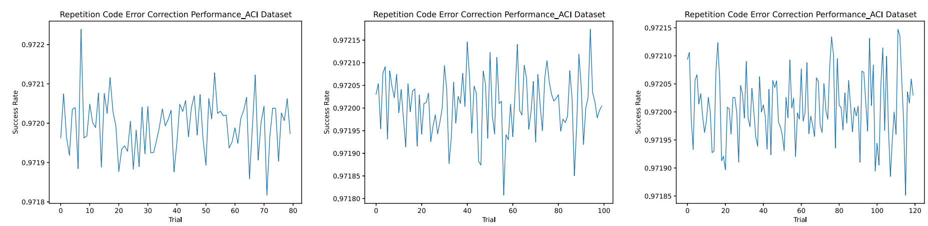


Figure 8 (a) Noise performance of BOT dataset using 80 trials (b) Noise performance of BOT dataset using 100 trials (c) Noise performance of BOT dataset using 120 trials

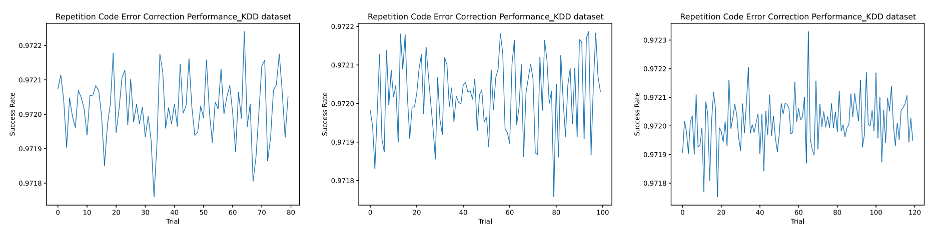
to the following Equation (7):

$$y_i = (w^T x_i - b) \geq 1, \quad \forall 1 < i < n \tag{7}$$



(a) (b) (c)

Figure 9 (a) Noise performance of ACI dataset using 80 trials (b) Noise performance of ACI dataset using 100 trials (c) Noise performance of ACI dataset using 120 trials



(a) (b) (c)

Figure 10 (a) Noise performance of KDD dataset using 80 trials (b) Noise performance of KDD dataset using 100 trials (c) Noise performance of KDD dataset using 120 trials

Here, y_i represents label, x_i represents features, b is bias and w is a vector to the hyper plane. Kernel is a fundamental component that reconstruct classical points into a new space, promoting smooth categorization of the dataset. However, straightforwardly scheming this transformation may be computationally harmful. Even with classical systems, it can still be computationally high-priced and inefficient to label important features appropriate for SVM. For enhanced feature mapping, quantum kernels harness the principles of quantum mechanics. These kernels translate classical binary bits into the quantum state space using qubits. In QSVMs, simple classical points are plan into a bigger-dimensional quantum feature utilizing a quantum feature map functions. This process is presented mathematically by a function $\phi(x)$, where x is the classic point and $\phi(x)$ is allure equivalent quantum feature $u_{\phi(x)}$. Once the data points are plan into the quantum features, quantum kernel function is used to measure the correspondence dot product between the quantum feature $\kappa(x_i, x_j)$ as given in the Equation (8):

$$\kappa(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2 \tag{8}$$

Then the ansatz (given in the Equation (9) and (10)) are used as a feature map.

$$u_{\phi(x)} = U_{\phi(x)} \bigotimes H^{\bigotimes n} \tag{9}$$

$$U_{\phi(x)} = exp(i \sum_{S \in n} \phi_S(x)) \prod_{i \in S} Z_i \tag{10}$$

For $S < 2$, the $\prod_{i\in S}$ will interact with $Z_i Z_j$ and the $\sum_{S\in n}$ will take n qubits. Then final circuit depth of 2 can be computed by using the Equation (11):

$$U_{\phi(x)} \bigotimes H^{\bigotimes n} U_{\phi(x)} \bigotimes H^{\bigotimes n} \tag{11}$$

Finally, the information regarding the quantum kernel is extracted from the quantum circuit in order to input it into the quantum SVM algorithm. The quantum kernel function computes the quantity calculated from two vectors of the quantity feature vectors $\phi(x_i)$ and $\phi(x_j)$. The quantum feature print and quantum kernel are usually executed as quantum circuits. These circuits manipulate quantum states to act the necessary renewals and computations. After the quantum kernel is computed for all pairs of data points, the QSVM optimizes a classical SVM objective function utilizing methods like classic growth algorithms or variational quantum algorithms.

4.8 Quantum Neural Networks (QNN)

The goal of quantum machine learning is to create novel methodologies by combining ideas from classical machine learning and quantum computing. This concept is embodied in QNNs, which combine parameterized quantum circuits and classical neural networks. QNN algorithm made to find hidden patterns in data, much as classical models. They have the ability to take classical data as input (x), transform it into quantum states $\phi(x)$. The summation is computed based on dot product of input x and weight w is given in the Equation (12):

$$\sum = x.w \tag{12}$$

In order to improve the learning, a constant value called bias (b) is added to the summation as given in the Equation (13):

$$k = \sum x.w + b \tag{13}$$

The activation function is applied to the final summation to produce the output y as given in the Equation (14):

$$y = \frac{1}{1 + e^{-k}} \tag{14}$$

The process of adjusting the weight is iterated until desired output is obtained or the error is minimal. The error is calculated as difference between actual and expected output as given in the Equation (15):

$$Error = Actual\ output - Expected\ output \tag{15}$$

The overall error is calculated by using the Equation (16):

$$Error = \frac{1}{n} \sum_{i=1}^{n} Actual\ output - Expected\ output \tag{16}$$

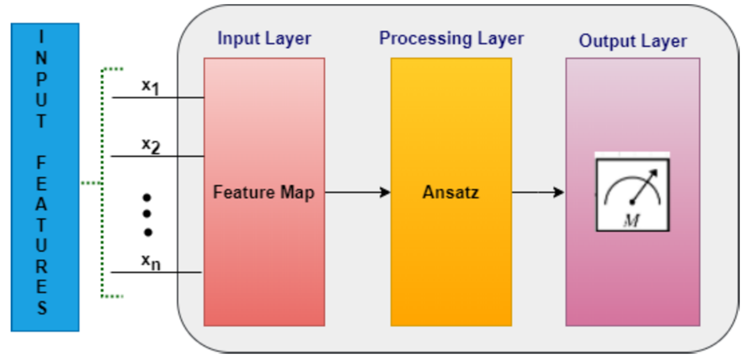


Figure 11 General architecture of a Quantum Neural Network, where input features are encoded through a feature map, processed by a quantum ansatz, and measured in the output layer for classification

Figure 11 represents the generic structure of QNN. In QNN, Anasatz function is used to adjust weight value for the given input features. The final output of quantum circuit is measured using the measurement unit.

For a given input features, $(|\phi(x_i)\rangle, y_i), i = 1 \dots n$ where $\phi(x_i)$ are input states y_i represents final outcome.

QNN is constructed using quantum circuits, that are sequences of quantum gates. Common quantum gate contains the Pauli gates (X, Y, Z), Hadamard gate (H), Phase gate (S) and Controlled-NOT gate $(CNOT)$. These gates are used to construct complication, superposition and perform different quantum operations. In QNN, quantum circuits may hold training parameters that are adjusted all along training to improve the network's performance. At the end of the quantum computation, qubits are calculated to get classical outcomes.

4.9 Hybrid Quantum Neural Network (HQNN)

Though some problems can be solved more quickly with quantum computing, only a small number of qubits and gate operations are currently possible due to technological limitations. Because of this, a hybrid approach is used in this work that combines quantum and conventional algorithms in a heterogeneous ensemble. In order to process large volume of data, the quantum techniques called Variational Quantum Circuits (VQCs) with classical deep learning are combined together for improved classification task. This hybrid approach, which makes use of both classical and quantum computing approaches. HQNN usually start with classical interconnected system layers, in the way that dense (sufficiently related), convolutional as shown in Fig. 12. The process involves extracting features, flattening them, and passing them through classical fully connected layers. Subsequently, these features are split into quantum layer processing, which is implemented as VQCs. Following the quantum processing stage, the output is measured using the measurement unit of the quantum circuit. This layer maps classical features into the quantum state of qubits. VQC consists of three primary parts: embedding, variational gates, and measurement. The "angle embedding" approach is used to accomplish the embedding. It modifies the rotation angles of qubits to encode classical properties. The input features are encoded as $|\phi\rangle = r_x|\phi(x_i)\rangle, i = 0 \dots n$, where r_x represents an angle of rotation applied on each input feature x. Each qubit, in its ground state, is rotated by an angle that is proportionate to the value that corresponds to the an angle in the input vector. The variational

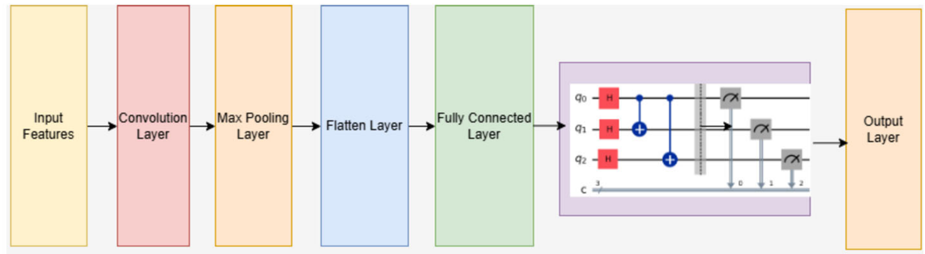


Figure 12 Hybrid quantum classical model architecture combining convolution, pooling, and fully connected layers with a quantum circuit layer, followed by measurement for final output classification

portion consists of a closed chain of CNOT operations after single-qubit rotations with trainable parameters for each qubit. While the CNOT gates entangle the qubits, the rotations alter the encoded input data in accordance with the variational parameters. Within each quantum layer, the repetitions of rotations and CNOT operations are defined by the depth of the variational components. Each individual quantum circuit has different variational parameters. Measurements made on the Pauli basis matrices comprise the measurement portion. The measurement operation (M) is given in the Equation (17):

$$M = \langle 0|r_x(\phi)u(\theta)y_i u(\theta)r_x(\phi)|0\rangle \tag{17}$$

where, y_i is the Pauli y matrix for the ith qubit and $r_x(\phi)$, $u(\theta)$ are the embedding operations applied over VQC circuits. These quantum circuits contain training parameters that are adjusted all along training to make better the network's performance. At the end of the computing, qubits in the VQCs are measured to acquire classical outcomes.

5 Results and discussion

in this section, the experimental setup and performance evaluation of the proposed method over real-time datasets are discussed. All experiments were performed on a workstation with an Intel Xeon E5620 CPU operating at 2.40 GHz, 15.6 GB RAM, and an NVIDIA TESLA GPU that accelerated the proposed method. The system was implemented using real-time datasets and executed on the widely accessible IBM Quantum platform.

First, the methodology was validated through quantum simulation, and then the experiments were deployed on an IBM quantum processor to assess real-device performance. For all datasets, 70% of the instances were used for training, while the remaining 30% were reserved for testing. Each dataset contains two classes of network traffic: Normal and Attack. The IBM Qiskit framework, which provides built-in quantum machine learning models, was utilized for implementing and evaluating the hybrid quantum–classical approach.

By applying dimensionality reduction on all the three datasets, essential features (shown in Table 2) are taken and trained with classical machine learning models as well as quantum machine leaning models given in Table 2. Moreover, Table 3 shows the detailed comparison results of datasets with classical ML and quantum ML models. For the BoT dataset, the classical machine learning models demonstrate moderate performance, with DT achieved a testing accuracy of 82.1%, the support vector machine (SVM) achieved

Table 2 Description of the selective features based on Principal Component Analysis for the benchmark datasets used in this study

Dataset	Features	Types	Description
BoT	HH_jit_L0.1_Variance	Continuous	Variance associated with HH_jit
	HH_jit_L0.1_mean	Continuous	Mean associated with HH_jit
	H_L0.1_variance	Continuous	Variance associated with H
	MI_dir_L0.1_variance	Continuous	Variance associated with MI_dir
	H_L0.1_weight	Continuous	Weight associated with H
ACI	Flow Duration	Continuous	Duration associated with Flow Duration traffic
	Flow IAT Total	Continuous	Total associated with Flow IAT flow
	Fwd IAT Max	Continuous	Max associated with Fwd IAT flow
	Flow IAT Max	Continuous	Max associated with Flow IAT flow
	Fwd IAT Mean	Continuous	Mean associated with Fwd IAT flow
KDD	dst_host_srv_count	Continuous	Number of connections to the same host and service
	srv_count	Continuous	Number of connections to the same service.
	count	Continuous	Number of connections from the same source
	duration	Continuous	Length of the connection
	src_bytes	Continuous	Bytes sent from source to destination.

78.4%, and the DNN achieved 91.2%. In comparison, the quantum learning models exhibit competitive performance, where the QSVM achieved 82.1%, the QNN reaches 83.5%, and the proposed Hybrid Quantum Neural Network (HQNN) attained the highest accuracy of 92.8%. Based on this comprehensive analysis, the HQNN clearly outperforms both classical and pure quantum models.

For the ACI dataset, the performance comparison clearly highlights the differences between classical and quantum learning models. The classical models namely DT achieved 90.3%, SVM obtained 88.1% and DNN model shown 91.2% which is noticeable improvement due to their increased learning capacity. The quantum models namely QSVM, QNN, and HQNN achieved 77.1%, 78.9%, and 92.8% as a testing accuracy respectively. However, the most significant improvement is observed in the proposed HQNN, which consistently achieves the highest accuracy among all evaluated techniques. For the KDD dataset, classical models namely DT achieved 90.1%, SVM obtained 94.1% and DNN model achieved 96.3% of testing accuracy. The quantum models namely QSVM, QNN, and HQNN achieved 90.9%, 95.6%, and 94.9% as a testing accuracy respectively.

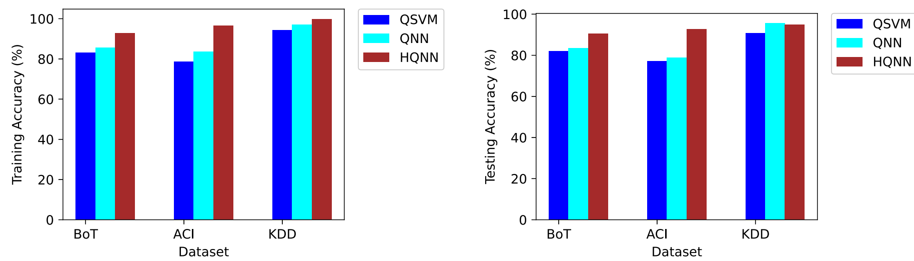
Figure 13(a) shows that, HQNN achieves the highest training accuracy of 92.8%, 96.7%, and 99.8% across all three datasets when compared to QNN and QSVM quantum models. Similarly the validation accuracy of proposed approach HQNN achieves 90.7%, 92.8%, and 94.9% for three datasets against QNN and QSVM is shown in Fig. 13(b).

Among the models considered, our proposed approach HQNN, achieved the lowest error of 0.124, 0.204, and 0.041 across all three datasets, surpassing other QNN and QSVM models as shown in Fig. 14(a).Since Quantum algorithms frequently entail important overhead on account of requirements like quantum state arrangement, measurement, and error. The HQNN can mitigate this overhead by utilizing quantum resources more seldom and capably. The HQNN can minimize computational time and resource overhead by focusing its processing on particular tasks within the network system architecture. Among the models considered, our proposed approach HQNN, achieves the highest F1-Score of 97.9%, 93.7%, and 98.5% across all the standard datasets, surpassing other QNN and QSVM models as shown in Fig. 14(b).

The proposed approach, HQNN, achieves the highest precision score of 97.1%, 95.7%, and 97.5% across all three datasets, surpassing other QNN and QSVM models as shown

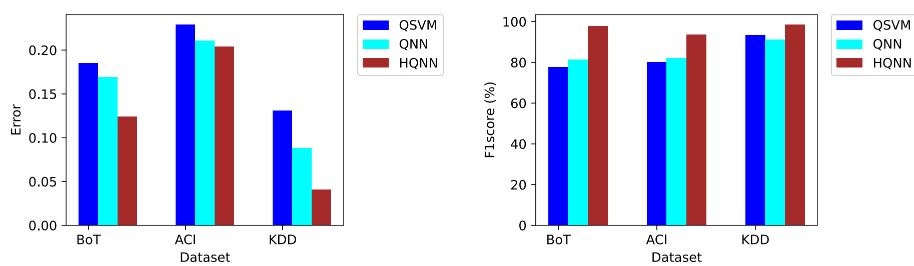
Table 3 Comparative performance analysis of classical machine learning models (DT, SVM, and DNN) and quantum machine learning models (QSVM, QNN, and HQNN) after applying PCA across three intrusion detection datasets

Dataset Name	Approaches	Algorithm	Training Accuracy	Testing Accuracy	Accuracy Error	Precision	Recall	F1 Score
BoT	Classical Approach	DT	0.907	0.821	0.179	0.862	0.717	0.783
		SVM	0.809	0.784	0.172	0.730	0.728	0.739
		DNN	0.912	0.901	0.068	0.915	0.926	0.920
	Quantum Approach	QSVM	0.831	0.821	0.185	0.827	0.731	0.776
		QNN	0.867	0.835	0.169	0.862	0.771	0.781
		HQNN	0.928	0.907	0.124	0.971	0.988	0.979
ACI	Classical Approach	DT	0.942	0.903	0.097	0.894	0.901	0.897
		SVM	0.918	0.881	0.119	0.861	0.873	0.867
		DNN	0.955	0.912	0.188	0.905	0.911	0.908
	Quantum Approach	QSVM	0.786	0.771	0.229	0.753	0.860	0.802
		QNN	0.837	0.789	0.211	0.756	0.902	0.822
		HQNN	0.967	0.928	0.204	0.957	0.918	0.937
KDD	Classical Approach	DT	0.992	0.901	0.090	0.870	0.890	0.910
		SVM	0.972	0.941	0.059	0.933	0.938	0.935
		DNN	0.995	0.963	0.037	0.957	0.961	0.959
	Quantum Approach	QSVM	0.945	0.909	0.131	0.931	0.939	0.934
		QNN	0.971	0.956	0.088	0.935	0.891	0.912
		HQNN	0.998	0.949	0.041	0.975	0.996	0.985



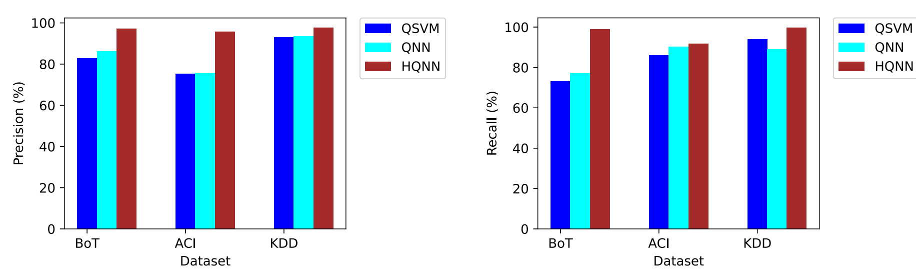
(a) (b)

Figure 13 (a) Training Accuracy of all the classifier (b) Validation Accuracy of all the classifier



(a) (b)

Figure 14 (a) Error analysis of various classifier (b) F1-score analysis of various classifier



(a) (b)

Figure 15 (a) Precision analysis of various classifier (b) Recall analysis of various classifier

in Fig. 15(a). Among the models considered, our proposed model, achieves the highest recall Score of 98.8%, 91.8%, and 99.6% across all three datasets, surpassing other QNN and QSVM models as shown in Fig. 15(b).

Figure 16(a), (b), (c) shows the comparison results of ROC performance with quantum implementation of HQNN, QNN and QSVM classifiers using the standard datasets. The proposed HQNN clearly demonstrates the supremacy of the quantum model over other quantum methods in terms of ROC. The Fig. 16 shows receiver operating characteristics (Roc) for BoT, ACI and KDD datasets in (a), (b), and (c) respectively.

The Fig. 17 depict the computational time taken on classical ML algorihtms, the simulator and the quantum computer for each dataset. It's evident that the time taken on the

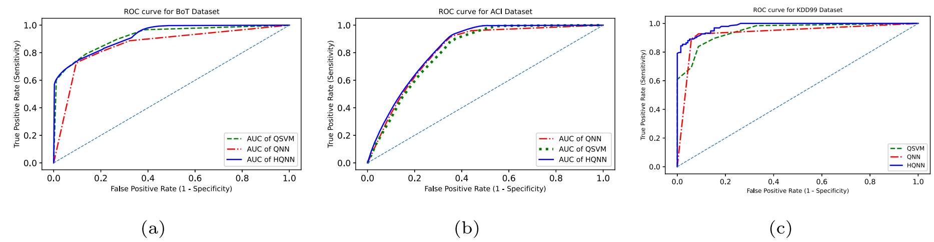


Figure 16 (a) ROC for the BoT dataset (b) ROC for the ACI dataset (c) ROC for the KDD dataset

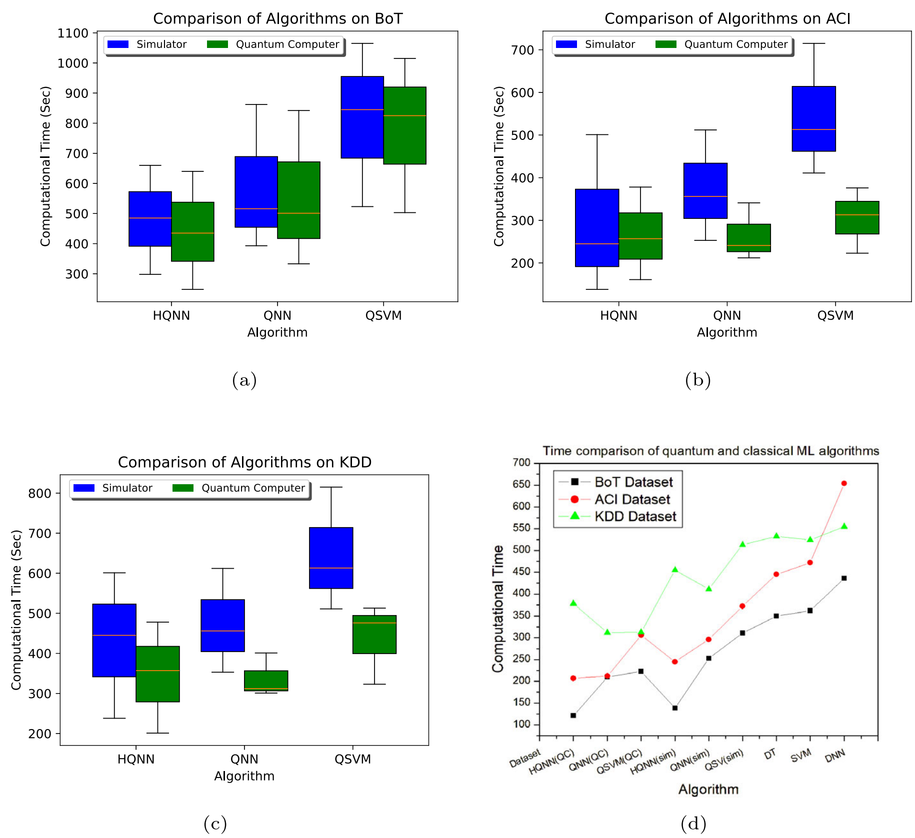


Figure 17 Computational time of (a) Quantum Variant models for IoT-BoT dataset executed on simulator and quantum platform (b) Quantum Variant models for ACI dataset executed on simulator and quantum platform (c) Quantum Variant models for KDD executed on simulator and quantum platform (d) Classical ML models on all three benchmark datasets

quantum computer is shorter compared to the simulator, and classical approaches. For the BoT dataset, the computation time of classical ML algorithms is compared with the proposed HQNN executed on both the quantum simulator and the quantum computer. From the Fig. 17, it can be observed that by executing HQNN in quantum simulator, it shows 65.42% improvement in training tome over the classical DT, 60.57% improvement when executed on the quantum computer. against the SVM model, the proposed approach on quantum simulator shows an improvement of 66.57% on the quantum simulator and

Table 4 Comparison of existing techniques across multiple IoT datasets and methodologies, along with the performance of the proposed HQNN model

Existing Work	Dataset	Method(s) Used	Accuracy
Isheeaq et al. [29]	BoT-IoT	Machine learning based IDS	90.34%
FL de Caldas et al. [30]	BoT-IoT	Feature selection and ML models for multilayer IoT attacks	89.75%
Rezakhani et al. [31]	ACI-IoT-2023	Few-Shot Learning	90%
Sudhakar et al. [4]	IDS	QML	92%
Sapre et al. [32]	KDDCup99	Comparison of multiple ML algorithms (SVM, RF, NB)	91.51%
Vinayakumar et al. [33]	KDDCup99	Deep learning based IDS	93%
Proposed Model	BoT-IoT	HQNN	90.70%
Proposed Model	ACI	HQNN	92.80%
Proposed Model	KDD	HQNN	94.90%

61.87% on the quantum computer. Similarly, for the DNN model, the computation of 72.24% improvement on the quantum simulator and 68.34% on the quantum computer.

For the ACI dataset, the computation time of classical ML algorithms is compared with the proposed HQNN executed on both the quantum simulator and the quantum computer. From the Fig. 17, it can be observed that HQNN achieves 53.48% improvement in training time over the classical DT model when executed on the simulator, and 44.94% improvement when executed on the quantum computer. For the SVM model, the computation time shows an improvement of 56.14% on the quantum simulator and 48.09% on the quantum computer. Similarly, for the DNN model, the computation time shows an improvement of 68.34% on the quantum simulator and 62.53% on the quantum computer.

For the KDD dataset, the computation time of classical ML algorithms is compared with the proposed HQNN executed on both the quantum simulator and the quantum computer. From the Fig. 17, it can be observed that HQNN achieves a 29.08% improvement in training time over the classical DT model when executed on the simulator, and a 14.63% improvement when executed on the quantum computer. For the SVM model, the computation time shows an improvement of 27.86% on the quantum simulator and 13.16% on the quantum computer. Similarly, for the DNN model, the computation time shows an improvement of 31.89% on the quantum simulator and 18.01% on the quantum computer.

The Table 4 presents a comprehensive comparison between state-of-the-art classical and quantum intrusion detection approaches and the proposed HQNN model, evaluated on standard benchmark datasets. The performance of the proposed HQNN can be attributed to its hybrid architecture, which effectively combines the expressive power of quantum feature embedding with the robust optimization capability of classical neural layers. Quantum circuits capture complex high-dimensional feature interactions that are difficult for classical models to represent efficiently, while the classical post-processing layers stabilize training and prevent quantum noise from dominating. This synergy enables HQNN to generalize better on real-world intrusion detection data. Therefore, the proposed HQNN approach demonstrates the best overall accuracy and learning efficiency, making it the most effective model among all the evaluated classical and quantum baselines for the three intrusion detection datasets.

6 Conclusion and future scope

This paper utilized PCA for selecting the most efficient subset of features from high-dimensional IoT traffic data. Major features from the first principal component were selected and then encoded using a quantum ZZFeatureMap, which utilizes repetition based encoding mechanisms are able to detect and mitigate errors arising during quantum processing. Further, the developed hybrid quantum neural network model is trained and evaluated those transformed features, thus efficiently learning over real-time network traces. Measured outputs through quantum circuit evaluations have shown that the proposed system performs with higher accuracy, lower error rates, and reduced computational time compared to conventional machine learning methods. This integration of PCA driven feature selection with quantum enhanced encoding and hybrid neural network architectures underlines the role of quantum machine learning in the computational challenge posed by large scale IoT intrusion detection. Employing the inherent parallelism of quantum computation, the HQNN framework proposed in this paper offers a scalable and highly effective mechanism for securing Industry 4.0 ecosystems IoT environments. As quantum technologies mature and hybrid computational frameworks evolve, models of this nature are expected to play a pivotal role in securing next-generation intelligent and interconnected systems.

Future research may investigate runtime strategies for dynamically allocating tasks between classical and quantum processors, depending on the complexity and dimensionality of incoming data streams. Extending the framework to integrate additional data modalities such as device behavior logs, sensor outputs, and contextual metadata may lead to more comprehensive and accurate intrusion detection.

Author contributions
1. Indira Bharathi has implemented the idea and drafting 2. Veeramani Sonai identified problems statement, idea formulation and compiling 3. Sridevi S involved in testing and validation of results

Funding information
Open access funding provided by Vellore Institute of Technology. The authors would like to thank and acknowledge the financial support provided under the "Meity Quantum Computer Applications Lab Cohort-2 funding scheme" by Meity, India, and AWS Braket service for quantum hardware implementation. The authors would like to thank Vellore Institute of Technology (VIT), Chennai.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable. The author has consented to the submission of the manuscript to the journal.

Competing interests
The authors declare no competing interests.

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Received: 5 April 2025 Accepted: 2 January 2026 Published online: 26 January 2026

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